

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Surface Areas and Volumes

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Surface area

1. Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius r is $3\pi r^2$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2025, 1 mark**

SOLUTION

A is true: the two flat circular faces fit together exactly and the curved surfaces make a complete sphere.

R is false: the surface area of a sphere is $4\pi r^2$. ($3\pi r^2$ is the total surface area of a solid hemisphere.)

Answer: (c)

2. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2

CBSE 2024, 1 mark

SOLUTION

Surface area of the sphere = $4\pi r^2$

Each solid hemisphere has total surface area $3\pi r^2$, so the two together have $6\pi r^2$.

Ratio = $4\pi r^2 : 6\pi r^2 = 2 : 3$

Answer: (c)

2 Volume

3. A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21}$ m³ of air. Find the height of the cylindrical part. Take $\pi = \frac{22}{7}$. **CBSE 2025, 3 marks**

SOLUTION

Let the radius be r m. Then the height of the cylinder is $2r$ m.

$$\text{Volume} = \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3$$

$$\frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{1408}{21}, \text{ so } \frac{176}{21}r^3 = \frac{1408}{21}$$

$r^3 = 8$, so $r = 2$ and the height of the cylinder is $2r = 4$.

Answer: 4 m

4. The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm **CBSE 2024, 1 mark**

SOLUTION

The largest cone has base diameter 2 cm and height 2 cm, so $r = 1$ cm and $h = 2$ cm.

$$\text{Volume} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 1 \times 2 = \frac{2\pi}{3} \text{ cu cm}$$

Answer: (d)

5. A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm³. Find the two radii. **CBSE 2024, 3 marks**

SOLUTION

Let the outer and inner radii be R and r . Then $R - r = 1$.

$$\text{Volume of metal} = \pi(R^2 - r^2)h: \frac{22}{7} \times (R + r)(R - r) \times 14 = 176$$

$$44(R + r) = 176, \text{ so } R + r = 4.$$

Solving with $R - r = 1$: $R = 2.5$ and $r = 1.5$.

Answer: outer radius 2.5 cm, inner radius 1.5 cm

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